

Classic Riddler

11 March 2022

Riddle:

We're playing a game where you have to pick four whole numbers. Then I will roll four fair dice. If any two of the dice add up to any one of the numbers you picked, then you win! Otherwise, you lose.

For example, suppose you picked the numbers 2, 3, 4 and 12, and the four dice came up 1, 2, 4 and 5. Then you'd win, because two of the dice (1 and 2) add up to at least one of the numbers you picked (3).

To maximize your chances of winning, which four numbers should you pick? And what are your chances of winning?

Solution:

I enumerate all number choices and dice rolls in the file `dice_sums.xlsx`. I use this to count up which rolls corresponded to successful choices. I find that the maximally successful choice is 4,6,8,10. The total probability of winning is $1,263/1,296 \approx 0.9753$.

By my reasoning, 4,6,8,10 must be the best choice because they are even numbers, and are centered around 7. 7 is the single most likely number to be a sum, with 6,8 and 5,9... increasingly less likely. So if only a single number had to be picked, 7 would be the best choice. But for more than one number to pick, it's better to pick numbers that are more spaced out (instead of just 5,6,7,8 for example). Picking even numbers specifically is good. With four dice, it's possible to roll all even or all odd numbers, and not have an odd sum at all, but it's not possible to avoid an even sum. So using four distinct even numbers centered around 7 maximizes the probability. In fact, if six numbers could be picked, 2,4,6,8,10,12 would guarantee a win.

So the solution is **4,6,8,10** with a winning probability of **$1,263/1,296$** .